

DAY-1

Student Name: Saloni Kumari

Branch: MCA

Subject Name: LB 3

UID: 25MCA20087

Section/Group: MCA-3

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Q1. A. Develop sparse matrix using linked list.

Code:

```
class Node:
    def __init__(self, r, c, val):
        self.row = r
        self.col = c
        self.val = val
        self.next = None

head = None

n = int(input("Enter number of non-zero elements: "))

for i in range(n):
    r = int(input("Row: "))
    c = int(input("Column: "))
    v = int(input("Value: "))

    new = Node(r, c, v)

    if head is None:
        head = new
    else:
        temp = head
        while temp.next:
            temp = temp.next
        temp.next = new

print("\nSparse Matrix (Linked List):")
print("Row\tCol\tValue")

temp = head
while temp:
    print(temp.row, "\t", temp.col, "\t", temp.val)
    temp = temp.next
```

Output:

```
PROBLEMS  OUTPUT  TERMINAL  DEBUG CONSOLE  PORTS  QUERY RESULTS

PS C:\Users\salon\OneDrive\Desktop\EST> python lg_cp1.py
Row: 2
Column: 2
Value: 8
Row: 3
Column: 3
Value: 9

Sparse Matrix (Linked List):
Row    Col    Value
1      1      6
2      2      8
3      3      9
PS C:\Users\salon\OneDrive\Desktop\EST> 
```

B. Traverse matrix in spiral manner.

Code:

```
rows = int(input("Enter number of rows: "))
cols = int(input("Enter number of columns: "))
```

```
matrix = []
```

```
print("Enter matrix elements:")
for i in range(rows):
    row = list(map(int, input().split()))
    matrix.append(row)
```

```
top, bottom = 0, rows - 1
left, right = 0, cols - 1
```

```
print("Spiral Traversal:")
```

```
while top <= bottom and left <= right:
```

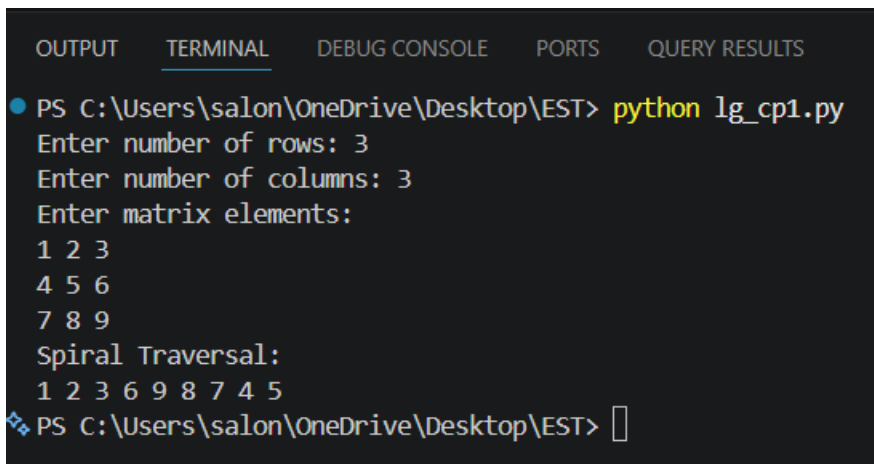
```
    # Left to Right
    for i in range(left, right + 1):
        print(matrix[top][i], end=" ")
    top += 1
```

```
    # Top to Bottom
    for i in range(top, bottom + 1):
        print(matrix[i][right], end=" ")
    right -= 1
```

```
    # Right to Left
    if top <= bottom:
```

```
for i in range(right, left - 1, -1):
    print(matrix[bottom][i], end=" ")
bottom -= 1
```

```
# Bottom to Top
if left <= right:
    for i in range(bottom, top - 1, -1):
        print(matrix[i][left], end=" ")
    left += 1
```

Output:

```
OUTPUT  TERMINAL  DEBUG CONSOLE  PORTS  QUERY RESULTS
PS C:\Users\salon\OneDrive\Desktop\EST> python lg_cp1.py
Enter number of rows: 3
Enter number of columns: 3
Enter matrix elements:
1 2 3
4 5 6
7 8 9
Spiral Traversal:
1 2 3 6 9 8 7 4 5
PS C:\Users\salon\OneDrive\Desktop\EST> 
```

Q2. Simulate a music playlist system using circular doubly linked lists.

Code:

```
class Node:
    def __init__(self, song):
        self.song = song
        self.next = None
        self.prev = None
```

```
head = None
```

```
def add_song(song):
    global head
    new = Node(song)
```

```
    if head is None:
        head = new
        new.next = new
        new.prev = new
    else:
        last = head.prev
        last.next = new
        new.prev = last
```

```
new.next = head
head.prev = new

def display():
    global head

    if head is None:
        print("Playlist is empty")
        return

    temp = head
    print("\nPlaylist:")
    while True:
        print(temp.song)
        temp = temp.next
        if temp == head:
            break

# Main Program
n = int(input("Enter number of songs: "))

for i in range(n):
    song = input("Enter song name: ")
    add_song(song)

display()
```

Output:

```
OUTPUT  TERMINAL  DEBUG CONSOLE  PORTS  QUERY RESULTS

PS C:\Users\salon\OneDrive\Desktop\EST> python lg_cp1.py
Enter song name: aankhe khuli ho ya ho bnd
Enter song name: kesariya
Enter song name: apna bna le
Enter song name: perfect

Playlist:
aankhe khuli ho ya ho bnd
kesariya
apna bna le
perfect
PS C:\Users\salon\OneDrive\Desktop\EST> |
```